William Ortega

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Education

Harvey Mudd College **Expected May 2028**

Bachelor of Science in Engineering, GPA: 3.4 *Claremont, CA*

- **Relevant Coursework:** Fluid Mechanics, Thermodynamics, Statics, Experimental Engineering, Design & Manufacturing, Signals & Systems, Differential Equations & Fourier Analysis, AI-Amplified Prototyping

Technical Skills

CAD & Simulation: SolidWorks, Ansys (CFD), FEA, COMSOL, OpenRocket
Programming: MATLAB, Python, C++, LabVIEW, SystemVerilog
Fabrication: 2-Axis Machining, 3D Printing, Laser & Waterjet Cutting, GD&T, DFM/DFA
Languages: English (Native), Spanish (Native)

Engineering Experience

Undergraduate Research Assistant, Flow Imaging Lab, Harvey Mudd College **Jan. 2026 – Present**

- Demonstrated higher thrust coefficients for submerged horizontal fins than surface-crossing fins (peak mean $C_T \approx 4.9$) by designing a horizontal-fin test fixture and running 100+ trials across fin thickness, amplitude, and frequency
- Characterized how aspect ratio, flapping frequency, and submersion depth affect thrust across 100+ trials with an ATI Mini40 six-axis load cell and NI DAQ, holding frequency within ±0.02 Hz
- Redesigned the lab's MATLAB thrust-processing pipeline for horizontal fins by remapping the thrust axis and rewriting the in-air inertial correction, enabling direct thrust-coefficient comparison with vertical fins

Propulsion Subteam Member, Mudd Amateur Rocketry Club (MARC) **Sept. 2026 – Present**

- Sized canards for the team's competition rocket with OpenRocket's optimizer, identifying dimensions that kept the stability margin above 1.2 calibers while preserving control authority
- Set up Ansys CFD simulations of the canards and fin tabs to build an aerodynamic lookup table vs. angle of attack for the rocket's flight controller

Engineering Projects

Autonomous Underwater Vehicle (Team of 4) **Jan. 2026 – May 2026**

- Built and deployed a PVC-frame AUV with proportional depth control that profiled light, pressure, and temperature off Dana Point, CA, measuring >80% surface-light loss by 3 m via Beer–Lambert analysis in MATLAB
- Designed and soldered a sensor/DAQ board integrating a TCS34725 RGB sensor (I^2C), MPX5700 pressure sensor, and NTC thermistor on a Teensy 4.1, logging depth-resolved data to microSD during motors-off holds to cut noise
- Designed a two-stage MCP6002 op-amp circuit (gain ≈ 6.87) that scaled the pressure signal into the 0.3–3.3 V ADC window to prevent clipping, and calibrated voltage-to-depth in saltwater with a linear fit

Mobile Lectern, Client Design Project (Team of 4) **Sept. 2025 – Nov. 2025**

- Designed and built a 3 lb, $24 collapsible lectern for a professor client that sets up in 1.6 s and holds over 12 lb
- Created the SolidWorks CAD model, drawings, and BOM, and laser-cut eco-friendly material for the final prototype

Machinist's Hammer **Sept. 2025 – Nov. 2025**

- Calibrated and operated manual mills and lathes to machine a hammer from raw steel and nylon to ±0.001 in.
- Applied drilling, tapping, heat treatment, and tempering while following technical drawings and a process router

Leadership & Activities

Co-Founder & Co-President, SHPE Harvey Mudd Chapter **Jan. 2026 – Present**

- Grew chapter engagement to 52 students through recruiting, outreach, and professional-development programming
- Secured $10K in institutional funding to send 16 members to the 2026 SHPE National Convention
- Planned 3 professional-development workshops and launched company outreach to build a sponsorship pipeline

President, Societal Advancement of Latinx (SALSA), Harvey Mudd College **Jan. 2025 – May 2026**

- Led and planned 5+ cultural and social events each semester to build community and belonging
- Expanded collaboration with five Latinx organizations across the Claremont Colleges